A three-dimensional counterexample to Lin's additive majorization conjecture

Complete solution of MathDB 332939

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Task 1. Resolve the all-power majorization conjecture. For positive semidefinite matrices A, B, let $|Z| = (Z^*Z)^{1/2}$ and write eigenvalues in decreasing order. Lin's Conjecture 4.3 asks whether

$$\lambda(A^2 + |BA|^p) \succ \lambda(A^2 + |AB|^p) \qquad \text{for every } p > 0.$$

Here $\succ$ is ordinary majorization. The answer is negative, already for $p = 4$ and real positive definite matrices of order three.

Take

$$A = \operatorname{diag}(1, 2, 20), \qquad B = \frac{1}{5}M, \qquad M = \begin{pmatrix} 6 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

The leading principal minors of M are $6, 11, 1$, so B is positive definite by Sylvester's criterion. The same is immediate for A. Set

$$L = A^2 + (AB^2A)^2 = A^2 + |BA|^4, \qquad R = A^2 + (BA^2B)^2 = A^2 + |AB|^4.$$

We shall certify

$$\lambda_{\min}(R) \le \frac{7027}{1250} < \frac{142}{25} < \lambda_{\min}(L). \tag{1}$$

Since L and R have equal trace, this contradicts $\lambda(L) \succ \lambda(R)$: the inequality for the two largest eigenvalues would require $\lambda_{\min}(L) \le \lambda_{\min}(R)$.

An exact lower bound for $\lambda_{\min}(L)$. Direct multiplication gives

$$X := AM^2A = \begin{pmatrix} 38 & -14 & 120 \\ -14 & 24 & 80 \\ 120 & 80 & 1200 \end{pmatrix}.$$

Consequently

$$T := 625L - 3550I = 625A^2 + X^2 - 3550I = \begin{pmatrix} 13115 & 8732 & 147440 \\ 8732 & 6122 & 96240 \\ 147440 & 96240 & 1707250 \end{pmatrix}.$$

The three leading principal minors of T are exactly

$$13115, \qquad 4042206, \qquad 152317908700.$$

They are positive, so T is positive definite. Hence $\lambda_{\min}(L) > 3550/625 = 142/25$.

An exact upper bound for $\lambda_{\min}(R)$. Put $v = (1, -1, 0)^{\mathsf{T}}$ and $Y = MA^2M$. Then

$$Yv = \begin{pmatrix} 54 \\ -31 \\ -5 \end{pmatrix}, \qquad 625R = 625A^2 + Y^2.$$

The Rayleigh quotient therefore gives

$$\lambda_{\min}(R) \le \frac{v^{\mathsf{T}}Rv}{v^{\mathsf{T}}v} = \frac{625 \cdot 5 + 54^2 + 31^2 + 5^2}{625 \cdot 2} = \frac{7027}{1250}.$$

The separation is strict because $142/25 - 7027/1250 = 73/1250 > 0$. Finally, cyclicity of the trace gives $\operatorname{Tr} X^2 = \operatorname{Tr} Y^2$; in fact $\operatorname{Tr}(625L) = \operatorname{Tr}(625R) = 1737137$. Thus (1) proves the failure of the required majorization. □

Relation to the cited even-power result. The earlier Corollary 2.3 of Lin and Wolkowicz proves an inequality involving the words $(AB)^k(BA)^k$ and $(BA)^k(AB)^k$. For $k > 1$ these are different from $|BA|^{2k}$ and $|AB|^{2k}$. Our counterexample concerns the latter expressions, precisely as in Conjecture 4.3. Ghabries's 2022 thesis [3], Remark 2.5, also records this additive majorization question after proving its determinantal consequence.

Verification. The accompanying script checks all displayed products, positivity certificates, trace identities and Rayleigh bounds over the rationals. All thirteen checks pass. The proof uses no numerical eigenvalues.

Task 2. Determine the valid exponent range. Identify all $p > 0$ for which the majorization statement holds for every pair of positive semidefinite matrices.

Task 3. Construct systematic counterexample families. Find parameterized positive definite pairs for which the least-eigenvalue inequality reverses, with explicit rational separation bounds.

Task 4. Identify structural sufficient conditions. Determine conditions beyond commutativity on A and B that guarantee the majorization for every positive exponent.

Task 5. Determine surviving spectral inequalities. Characterize which convex spectral functionals retain the asserted comparison even when full additive majorization fails.

References

[1] M. Lin, *On a determinantal inequality arising from diffusion tensor imaging*, Communications in Contemporary Mathematics **19** (2017), 1650044. Conjecture 4.3 in arXiv:1604.04141.

[2] M. Lin and H. Wolkowicz, *An eigenvalue majorization inequality for positive semidefinite block matrices*, Linear and Multilinear Algebra **60** (2012), 1365–1368. Corollary 2.3 in the author's final manuscript.

[3] M. M. Ghabries, *Contributions to Matrix Inequalities and Some Applications*, doctoral thesis, Université d'Angers, 2022, Remark 2.5, p. 50, author-uploaded full text.